Chapter 7. Logit Models for Multinomial Responses

Baseline-category and cumulative logits

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7.1 Nominal responses: baseline-category logit models

Motivation. Pairing each of J unordered categories with a baseline gives $J-1$ logit equations that determine all pairwise odds.

Let Y have J categories, $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$; at fixed $\mathbf{x}$ the counts are multinomial. Any suitable $J-1$ of the $\binom{J}{2}$ pairwise logits determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. So the baseline is arbitrary: changing it reparametrizes, not changing the model or fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention covers a or $b = J$). Each parametrization is a linear reparametrization of the others, so the set of probability vectors is the same. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; otherwise they still compare nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

219 alligators, four Florida lakes; response: primary food type (fish, invertebrate, reptile, bird, other); predictors lake L, gender G, size S (≤ 2.3 m vs. > 2.3 m). Sparse (219 in 80 cells): G^2 suits model comparison better than fit testing. $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$ ($df = 4$ each): collapse over gender. Then $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), versus 38.2 for (L) and 66.2 for (S); both matter. Counts (fitted for $L+S$):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Two of 40 standardized residuals exceed 2. Fish baseline; $s = 1$ for small; z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George reference). Estimates (SE):

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large ones (Wald CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$); these odds are higher in Trafford and Oklawaha, lower in Hancock, than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

Baseline parameters are zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$ leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. For quantitative x, plot $\hat{\pi}_j(x)$; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone.

7.2 Ordinal responses: cumulative logit models

Motivation. Using the ordering gives parsimony and power; cumulative logits with a common effect, from a latent continuous response, are the standard choice.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; alone it is a binary logit for $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.5 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^{β} is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every cutpoint: McCullagh's *proportional odds*. The curves for $j = 1, 2, 3$ ($J = 4$) have the same shape (Figure 1).

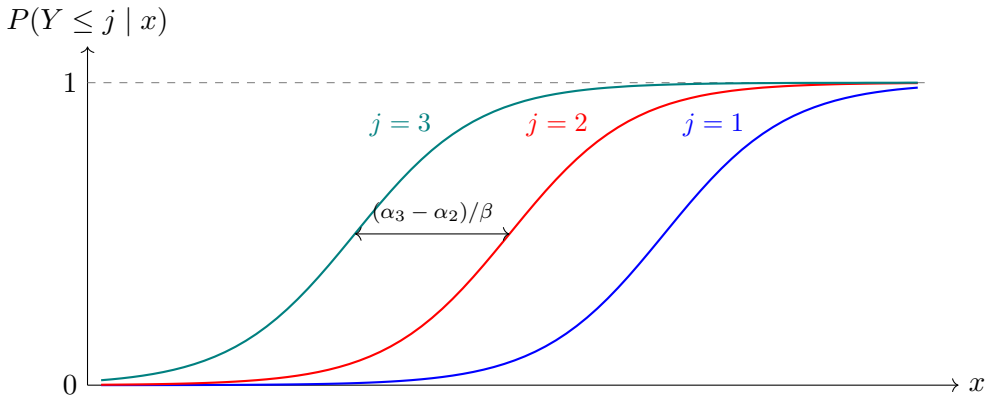


Figure 1: Proportional odds, $J = 4$, $\beta > 0$: three logistic curves with a common slope, horizontally shifted.

The shared shape makes the fit differ from separate binary fits. With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n}\prod_{j=1}^{J}\Big[\frac{\exp(\alpha_j + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}{1+\exp(\alpha_j + \boldsymbol{\beta}^{\top}\mathbf{x}_i)} - \frac{\exp(\alpha_{j-1} + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}{1+\exp(\alpha_{j-1} + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}\Big]^{y_{ij}} \tag{7.4}$$

with $\alpha_0 = -\infty$, $\alpha_J = +\infty$; it is maximized by Fisher scoring.

7.2.3 Mental impairment example

Example 7.6 (Mental impairment)

40 adults, Alachua County, Florida. Y = mental impairment (1 well, 2 mild, 3 moderate symptom formation, 4 impaired); x_1 = life events index (past three years); x_2 = SES (1 high, 0 low). (x_2, x_1) by response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	−0.282	0.642	(−1.56, 0.98)
α_2	1.213	0.661	(−0.05, 2.57)
α_3	2.209	0.721	(0.86, 3.71)
life (β_1)	−0.319	0.121	(−0.57, −0.09)
SES (β_2)	1.111	0.611	(−0.06, 2.35)

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y=1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1+\exp[-0.282 - 0.319(4.275)]} = 0.16.$$

$P(Y \le j)$ decreases with life events and is higher at high SES: given life events, the cumulative odds at high SES are $e^{1.111} = 3.0$ times those at low SES. At $\bar{x}_1$, $\hat{P}(Y=1) = 0.37$ (high) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high) and 0.28 to 0.09 (low). Score test of proportional odds: 2.33, $df = 4$, $P = 0.68$.

7.2.4 More complex models

Interactions and factors enter as in binary logistic regression, with $J-1$ cutpoint intercepts. Adding the interaction:

$$\operatorname{logit} \hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which cross for some $\mathbf{x}$ (misordered cumulative probabilities). A small P-value in the proportional odds test may reflect large n rather than important differences, and the simpler model may have smaller MSE. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters;
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument allows any inverse-cdf link, such as probit or complementary log-log.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

Model (7.5) arises from crude measurement of a latent Y^* with cdf $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on the latent scale ($\pm\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$, normal Y^*): in $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k SDs of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, extreme-value latent): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$, the grouped-time *proportional hazards* model; $P(Y \le j)$ approaches 1 faster than 0. The log-log link is cloglog for reversed categories ($G \mapsto 1 - G(-z)$).

Proposition 7.7 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs, fitted by McCullagh's Fisher scoring on (7.4) with G for the logistic cdf. For large n the maximum is unique; the log likelihood is concave for logit, probit and cloglog links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.8 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ones, suggesting cloglog (also implied by a Gompertz hazard). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat\beta_1 = -0.658,\ \hat\beta_2 = 0.626.$$

(These are population values.) By Proposition 7.7, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival beyond any age is women's to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. So if white women survive with probability π, white men and black women do so with about π^2, black men with about π^4.

7.4 Selected exercises

Motivation. Theory problems from Agresti's chapter end, filling in structural facts used above.

Exercise 7.24 (Multinomial trial as an exponential family)

For one multinomial trial with probabilities $\pi_1, \dots, \pi_J > 0$, let $\mathbf{y} = (y_1, \dots, y_{J-1})^\top$, $y_j = \mathbb{1}\{\text{category } j\}$. Show that $\mathbf{y}$ has the form

$$f(\mathbf{y}; \boldsymbol{\theta}) = \exp[\mathbf{y}^\top \boldsymbol{\theta} - b(\boldsymbol{\theta})],$$

a $(J-1)$-parameter exponential family whose natural parameters are the baseline-category logits $\theta_j = \log(\pi_j / \pi_J)$. Find b and check that $\partial b / \partial \theta_j = \mathbb{E}(y_j)$.

Solution.

Step 1. Put $y_J = 1 - \sum_{j<J} y_j$. Exactly one of $y_1, \dots, y_J$ equals 1, so $P(\mathbf{y}) = \prod_{j=1}^{J} \pi_j^{y_j}$.

Step 2. Factor out π_J:

$$\prod_{j=1}^{J} \pi_j^{y_j} = \pi_J^{1-\sum_{j<J} y_j} \prod_{j<J} \pi_j^{y_j} = \pi_J \prod_{j<J} \Big(\frac{\pi_j}{\pi_J}\Big)^{y_j} = \exp\Big[\sum_{j<J} y_j \theta_j + \log \pi_J\Big].$$

Step 3. By Proposition 7.3 (with no predictors), $\pi_J = 1/(1 + \sum_{h<J} e^{\theta_h})$, so $-\log \pi_J = b(\boldsymbol{\theta}) := \log(1 + \sum_{h<J} e^{\theta_h})$, a function of $\boldsymbol{\theta}$ alone. This gives the stated form with dispersion $a(\phi) = 1$ and $c(\mathbf{y}, \phi) = 0$; the map $(\pi_1, \dots, \pi_{J-1}) \mapsto \boldsymbol{\theta}$ is one-to-one onto $\mathbb{R}^{J-1}$.

Step 4. $\partial b / \partial \theta_j = e^{\theta_j}/(1 + \sum_h e^{\theta_h}) = \pi_j = \mathbb{E}(y_j)$, as for any exponential family.

So (7.1) uses the canonical link of this multivariate family, the analogue of logistic regression as a canonical GLM, with sufficient statistics $\sum_i y_{ij}$ and $\sum_i x_{ik} y_{ij}$. ■

Exercise 7.27 (A nonmonotone category probability)

Let $J = 3$ and

$$\pi_3(x) = \frac{1}{1 + \exp(\alpha_1 + \beta_1 x) + \exp(\alpha_2 + \beta_2 x)}.$$

Show that $\pi_3(x)$ is
(a) decreasing in x if $\beta_1 > 0$ and $\beta_2 > 0$;
(b) increasing in x if $\beta_1 < 0$ and $\beta_2 < 0$;
(c) nonmonotone when β_1 and β_2 have different signs.

Solution. Let $D(x) = 1 + e^{\alpha_1 + \beta_1 x} + e^{\alpha_2 + \beta_2 x} > 1$, so $\pi_3 = 1/D$ and $\pi_3' = -D'/D^2$ with

$$D'(x) = \beta_1 e^{\alpha_1 + \beta_1 x} + \beta_2 e^{\alpha_2 + \beta_2 x}.$$

(a) If $\beta_1, \beta_2 > 0$ then $D' > 0$, so $\pi_3' < 0$ everywhere.

(b) If $\beta_1, \beta_2 < 0$ then $D' < 0$, so $\pi_3' > 0$ everywhere.

(c) Say $\beta_1 > 0 > \beta_2$ (the other case is symmetric).

Step 1. $D''(x) = \beta_1^2 e^{\alpha_1 + \beta_1 x} + \beta_2^2 e^{\alpha_2 + \beta_2 x} > 0$, so D' is strictly increasing.

Step 2. As $x \to +\infty$ the first term of D' tends to $+\infty$ and the second to 0; as $x \to -\infty$ the first tends to 0 and the second to $-\infty$. Hence D' changes sign exactly once, at some x_0.

Step 3. Therefore $\pi_3' > 0$ for $x < x_0$ and $\pi_3' < 0$ for $x > x_0$: π_3 rises to a maximum at x_0 and then falls (with $\pi_3 \to 0$ at both ends). In particular it is not monotone.

This is the nonmonotonicity noted after the alligator example. ■

Exercise 7.31 (Cumulative logits need a common effect)

Consider logit $P(Y \le j \mid x) = \alpha_j + \beta_j x$, $j = 1, \dots, J-1$, with x ranging over $\mathbb{R}$ and the β_j not all equal. Show that the cumulative probabilities are misordered for some x, so the model does not define a probability distribution for all x.

Solution.

Step 1. Since the β_j are not all equal, there is a j with $\beta_j \neq \beta_{j+1}$ (otherwise all consecutive slopes, hence all slopes, would coincide).

Step 2. The difference of the two logits,

$$g(x) = (\alpha_j + \beta_j x) - (\alpha_{j+1} + \beta_{j+1} x) = (\alpha_j - \alpha_{j+1}) + (\beta_j - \beta_{j+1}) x,$$

is a nonconstant linear function, so $g(x) > 0$ for all x on one side of the root $x^* = (\alpha_{j+1} - \alpha_j)/(\beta_j - \beta_{j+1})$.

Step 3. The logistic cdf F is strictly increasing, so for such x, $P(Y \le j \mid x) = F(\alpha_j + \beta_j x) > F(\alpha_{j+1} + \beta_{j+1} x) = P(Y \le j+1 \mid x)$. This would make $\pi_{j+1}(x) = P(Y \le j+1 \mid x) - P(Y \le j \mid x) < 0$, which is impossible.

The two logistic curves have different slopes, so they cross (the remark in Section 7.2.4). On a bounded range of x the model can still be valid. ■

Exercise 7.32 (Uniform association in cumulative odds ratios)

For an $I \times J$ table with ordinal Y, independent multinomial rows, and scores $x_i = i$ for the rows, consider

$$\text{logit}\, P(Y \le j \mid X = x_i) = \alpha_j + \beta x_i.$$

(a) Show that $\text{logit}\, P(Y \le j \mid X = x_{i+1}) - \text{logit}\, P(Y \le j \mid X = x_i) = \beta$, and that this is the log cumulative odds ratio of the 2×2 table formed by rows $i, i+1$ and the binary response $(Y \le j,\ Y > j)$.
(b) Show that the residual df equal $IJ - I - J$.
(c) Show that independence of X and Y is the special case $\beta = 0$.
(d) If βx_i is replaced by unordered row parameters μ_i (treating X as nominal), show that for rows a, b the log cumulative odds ratio equals $\mu_a - \mu_b$ for all $J - 1$ cutpoints.

Solution.

(a) The difference is $(\alpha_j + \beta(i+1)) - (\alpha_j + \beta i) = \beta$. Writing $P_i = P(Y \le j \mid X = x_i)$, the logit difference is

$$\log \frac{P_{i+1}/(1 - P_{i+1})}{P_i/(1 - P_i)},$$

the log odds ratio of the 2×2 table with rows $i+1, i$ and columns $Y \le j$, $Y > j$. It equals β for all adjacent rows and cutpoints: uniform association in cumulative odds ratios.

(b) The saturated model has $I(J-1)$ parameters, the model J ($J-1$ intercepts and β); residual df $I(J-1) - J = IJ - I - J$.

(c) If $\beta = 0$, $P(Y \le j \mid X = x_i) = F(\alpha_j)$ is free of i: independence. Conversely, independence requires $\alpha_j + \beta x_i$ free of i, i.e. $\beta = 0$. So testing $\beta = 0$ is a one-df test of independence using both orderings.

(d) With $\text{logit}\, P(Y \le j \mid X = a) = \alpha_j + \mu_a$, the log cumulative odds ratio of rows a, b is $(\alpha_j + \mu_a) - (\alpha_j + \mu_b) = \mu_a - \mu_b$, free of j; the rows are stochastically ordered by μ_i (Proposition 7.5(c)). ■